Network Connectivity Testing

Network Topology

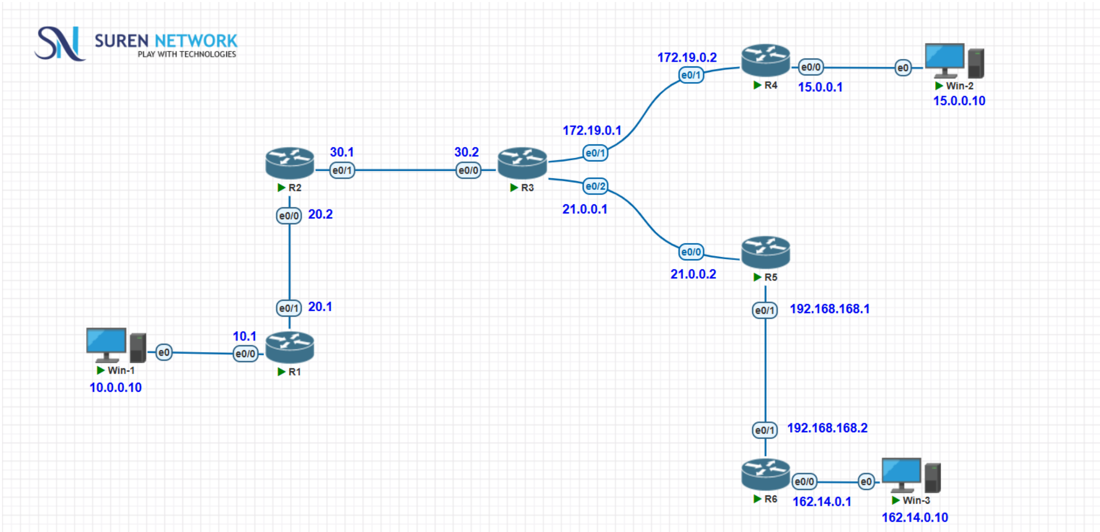


Figure 1: Network topology used for the connectivity testing lab.

This topology contains Win-1, Win-2 and Win-3 connected through routers R1 to R6. The following steps verify end-to-end connectivity between the Windows machines using ICMP ping.

1.Objective

To verify end-to-end network connectivity between the Windows virtual machines using the ping command

2. IP Addressing

Device	Interface	IP Address
Win-1	e0	10.0.0.10
R1	e0/0	10.0.0.1
R1	e0/1	20.0.0.1
R2	e0/0	20.0.0.2
R2	e0/1	30.0.0.1
R3	e0/0	30.0.0.2
R3	e0/1	172.19.0.1
R3	e0/2	21.0.0.1
R4	e0/1	172.19.0.2
R4	e0/0	15.0.0.1
Win-2	e0	15.0.0.10
R5	e0/0	21.0.0.2
R5	e0/1	192.168.168.1
R6	e0/1	192.168.168.2
R6	e0/0	162.14.0.1
Win-3	e0	162.14.0.10

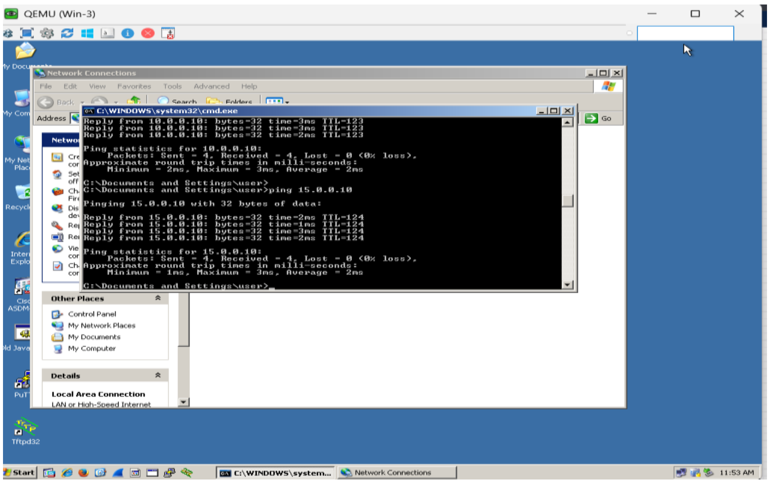
Step 1 – Open Command Prompt on Win-3

Start Win-3 and open Command Prompt using Start - Run - type cmd - Enter.

Step 2 – Test Win-3 to Win-2

Run:

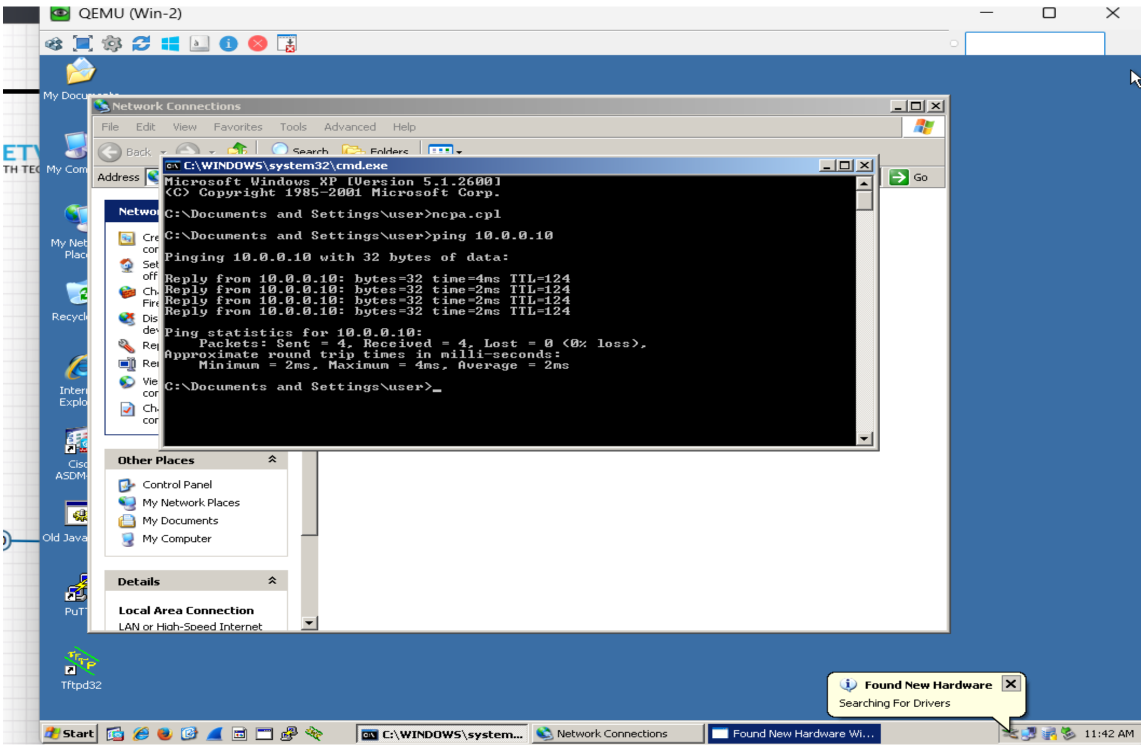
ping 15.0.0.10 Successful replies and 0% packet loss confirm that Win-3 can reach Win-2.



Step 4 – Test Win-2 to Win-1

On Win-2, open Command Prompt and run

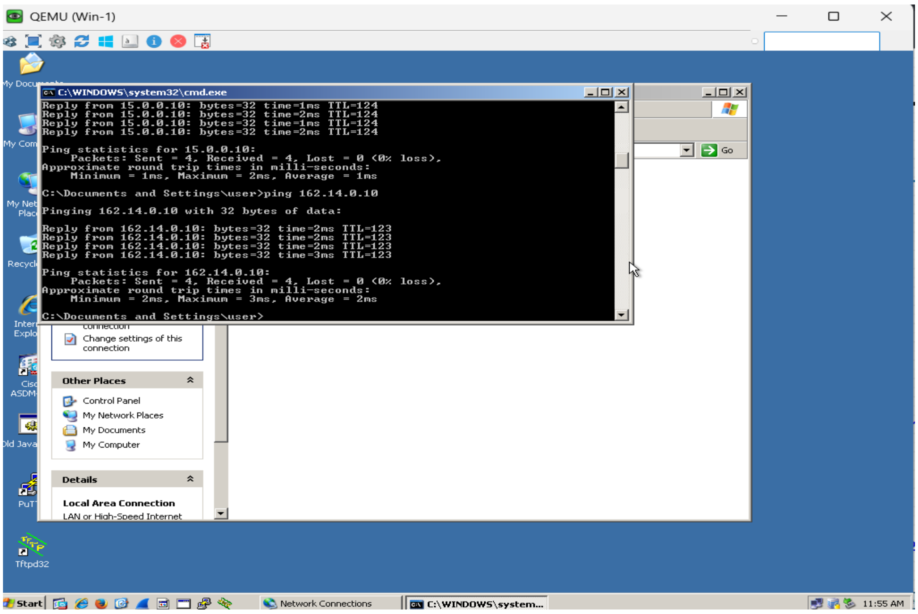
ping 10.0.0.10 The test shows successful replies from Win-1 with 0% packet loss.



Step 5 - Test from Win-1

On Win-1, test Win-3 and Win-2 using their IP addresses:

ping 162.14.0.10 , ping 15.0.0.10,



Verify the Results

Source	Destination	Result
Win-3	10.0.0.10 (Win-1)	Successful / 0% loss
Win-3	15.0.0.10 (Win-2)	Successful / 0% loss
Win-2	10.0.0.10 (Win-1)	Successful / 0% loss
Win-1	162.14.0.10 (Win-3)	Successful / 0% loss
Win-1	15.0.0.10 (Win-2)	Successful / 0% loss

Conclusion

The connectivity tests were completed successfully. The Windows machines received ICMP replies with 0% packet loss, confirming that the configured IP addressing and routing paths are functioning correctly.